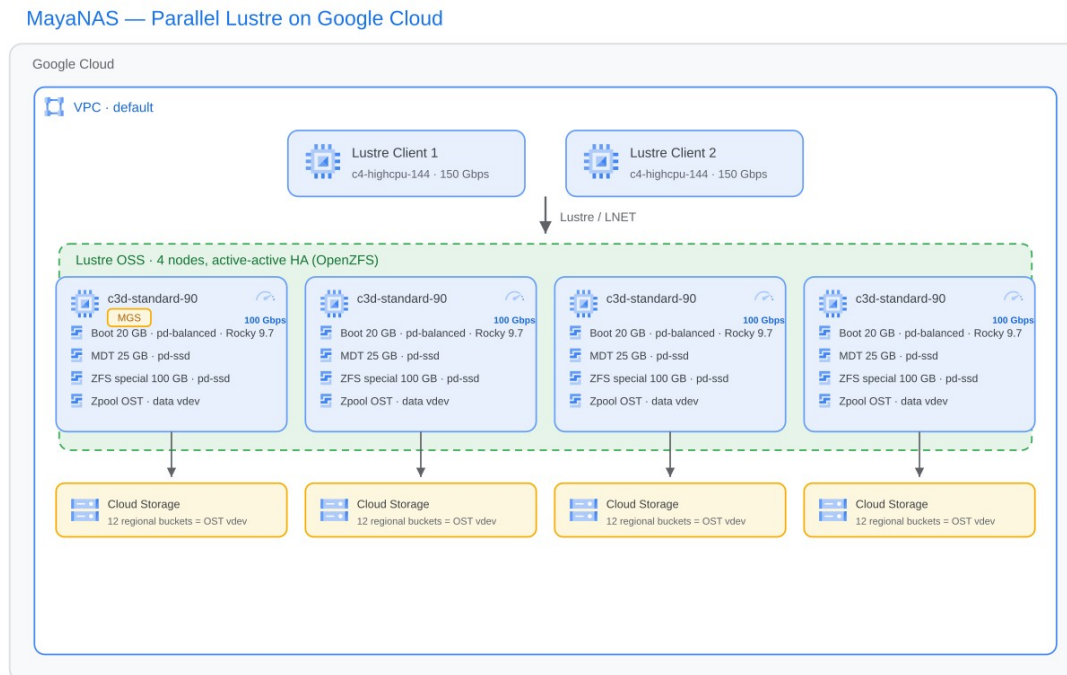


MayaNAS Lustre — MLPerf Storage v3.0 (Checkpointing 70B)

Submitter: ZettaLane Systems · **Division:** Closed · **System:** mayanas-lustre-gcp_c4hc144_c3d90_2c_4s

architecture



MayaNAS Storage Cluster (4 nodes, active-active HA)

- 4 × GCP c3d-standard-90 (AMD EPYC 9B14 Turin, 90 vCPU / 45 cores, 354 GB RAM)
- Rocky Linux 9.7 · 100 Gbps Tier_1 networking
- Each node: Lustre OSS with 1 OST (OpenZFS zpool; data vdev = 12 Standard regional GCS buckets via objbacker.io)
- Per node: Boot 20 GB pd-balanced · MDT 25 GB pd-ssd · ZFS special 100 GB pd-ssd
- MGS on node 1's VIP · regional buckets enable HA failover (surviving node imports partner zpool)

Benchmark Client

- 2 × GCP c4-highcpu-144 (Intel Xeon 6985P-C (Granite Rapids), 144 vCPU, 283 GB RAM)
- Ubuntu 24.04 · 150 Gbps networking
- Lustre client mount (hybrid_io=1), distributed MPI 2 hosts × 32 ranks

Results

- Checkpointing (llama3-70b, np=64 = 2×32): save ~32.4 GB/s, load ~21.2 GB/s (Pattern A, 10W + 10R via MPI)

Availability

MayaNAS and MayaScale can be deployed from the Google Cloud Marketplace portal or as Terraform modules (<https://github.com/zettalane-systems/open-lustre-cloud>). Contact: info@zettalane.com.